

Reaction Mechanism of the PET Degrading Enzyme PETase Studied with DFT/MM Molecular Dynamics Simulations

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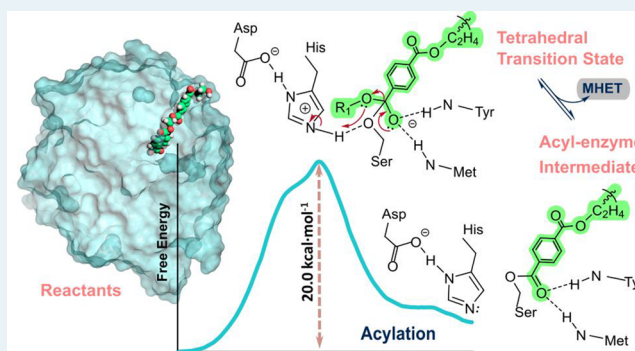
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ABSTRACT: Polyethylene terephthalate (PET) has been widely used to make disposable bottles, among others, leading to massive PET waste accumulation in the environment. The discovery of the *Ideonella sakaiensis* PETase and MHETase enzymes, which hydrolyze PET into its constituting monomers, opened the possibility of a promising route for PET biorecycling. We describe an atomistic and thermodynamic interpretation of the catalytic reaction mechanism of PETase using umbrella sampling simulations at the robust PBE/MM MD level with a large QM region. The reaction mechanism takes place in two stages, acylation and deacylation, each of which occurs through a single, associative, concerted and asynchronous step. Acylation consists of proton transfer from Ser131 to His208, concerted with a nucleophilic attack by Ser131 on the substrate, leading to a tetrahedral transition state, which subsequently results in the release of MHET after the breaking of the ester bond. Deacylation is driven by deprotonation of an active site water molecule by His208, with the resulting hydroxide attacking the acylated Ser131 intermediate and breaking its bond to the substrate. Subsequently, His208 transfers the water proton to Ser131, with ensuing formation of MHET and enzyme regeneration. The rate-limiting acylation has a free energy barrier of 20.0 kcal·mol⁻¹, consistent with the range of experimental values of 18.0–18.7 kcal·mol⁻¹. Finally, we identify residues whose mutation should increase the enzyme turnover. Specifically, mutation of Asp83, Asp89, and Asp157 by nonpositive residues is expected to decrease the barrier of the rate-limiting step. This work led to the understanding of the catalytic mechanism of PETase and opened the way for additional rational enzyme engineering.

KEYWORDS: plastic accumulation, biodegradation, biorecycling, QM/MM, density functional theory, umbrella sampling



1. INTRODUCTION

Plastics have become an integral part of our lives due to their resistance, durability, and low cost, resulting in their wide use and accumulation as waste in terrestrial and marine environments, with severe environmental and ecological consequences.¹ The production of plastics worldwide has increased exponentially since 1950, with 359 million tons produced in 2018 only.² According to the United Nations Development Programme, plastic pollution threatens our ecosystems, biodiversity, and health, at a pace and scale never seen before. It is estimated that one million plastic drinking bottles are purchased every minute, and up to five trillion disposable plastic bags are used every year.³

Single-use plastics, such as polyethylene terephthalate (PET), are among the materials most used as containers for liquid drinks and food, and are commonly discarded afterward.⁴ PET is obtained through a polycondensation reaction between terephthalic acid (TPA) and ethylene glycol (EG) or by transesterification of dimethyl terephthalate (DMT) and EG (Scheme 1).

PET is a semicrystalline, thermoplastic polyester whose ester groups confer superior resistance to biodegradation. Consequently, PET is a major component of plastic waste in the environment, together with polyethylene (PE), polypropylene (PP), and polystyrene (PS).^{6–8}

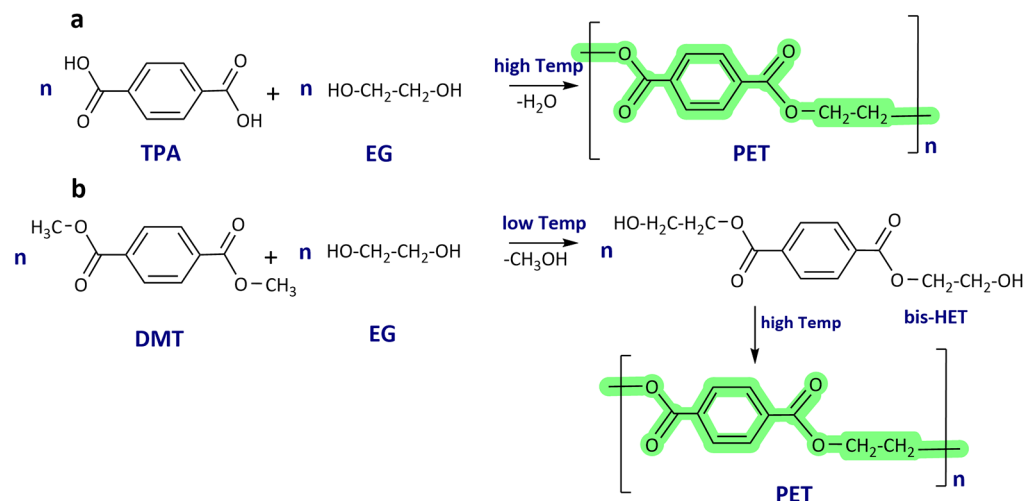
Removal of PET waste includes landfill, incineration, and recycling. The first two processes are far from being a solution, resulting in other environmental problems, namely groundwater pollution or CO₂ emissions.⁶ The recycled PET is reprocessed to films, sheets, or textile fibers, but this process requires heat, and consequently, the PET mechanical properties get lost. Alternatively, chemical recycling is not economically feasible because the reprocessed resin is more expensive than fossil fuel monomers.⁷ The biodegradation of

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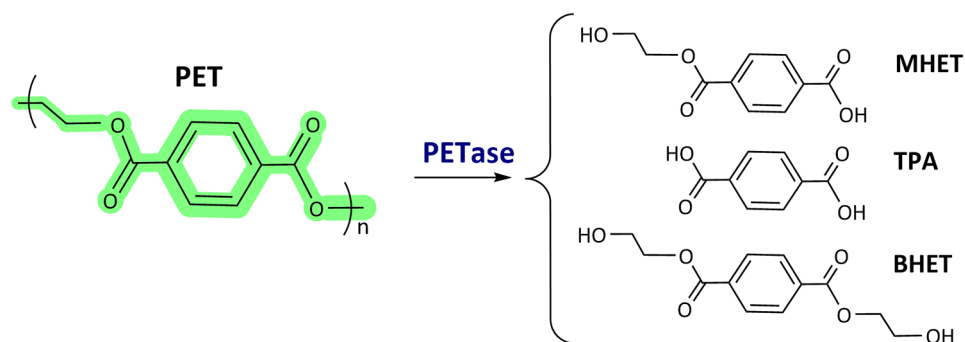
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Scheme 1. Synthesis of PET^a

^aa. Polycondensation of PET using terephthalic acid (TPA) and ethylene glycol (EG). The reaction occurs at high temperatures and water is eliminated. b. Polycondensation of PET through transesterification. This reaction takes place in two steps: the first, the transesterification of dimethyl terephthalate (DMT) with ethylene glycol (EG) at low temperature, with the formation of bis(2-hydroxyethyl) terephthalate (bis-HET) and methanol elimination. The second step is the polycondensation of bis-HET at high temperatures.⁵

Scheme 2. PET Degradation^a

^aPETase catalyzes the depolymerization of PET to mono (2-hydroxyethyl) terephthalate (MHET), terephthalic acid (TPA), and bis(2-hydroxyethyl) terephthalate (BHET).

plastics has become a widely desired solution due to its ecological nature and affordability.¹ Several bacterial enzymes acting on polyester degradation have been studied and described in recent years, including cutinases, lipases, and esterases.^{9,10} The enzymatic hydrolysis of synthetic polyesters like PET has been proposed as a more efficient and environmentally friendly method for recycling plastic waste.¹⁰

Considerable advance was made with the research led by Shosuke Yoshida and collaborators, in which a new species of bacteria was discovered, *Ideonella sakaiensis* 201-F6, capable of using PET as a source of carbon and energy. This bacterium has two hydrolases, PETase and MHETase, that together convert PET back into terephthalic acid and ethylene glycol. PETase carries out the most challenging work of PET biodegradation, which is the hydrolysis of PET to mono (2-hydroxyethyl) terephthalate (MHET), with small traces of terephthalic acid (TPA) and bis(2-hydroxyethyl) terephthalate (BHET;^{11–14} Scheme 2).

PETase can degrade other semiaromatic polyesters, e.g., polyethylene-2,5-furandicarboxylate (PEF),¹² making it a valuable target for the biodegradation of other plastics. Biodegradation in the environment of recalcitrant plastics

such as PET is a complex process that results from the combination of biotic and abiotic factors. The hydrophobicity, degree of crystallinity, surface topography and molecular size of synthetic polymers, and polymer chain mobility are factors that restrict the attack of microorganisms to plastic products. In particular, PET contains aromatic terephthalate units that reduce the chain mobility, resulting in extremely low hydrolyzability of the backbone ester linkages. Since PET is a semicrystalline plastic, the enzymatic attack will be easier in the amorphous domain, so the biodegradation rate decreases with increasing crystallinity.^{8,15} However, as one of the goals of PET biodegradation is to make the process viable at higher temperature (>60 °C), where PET is amorphous, the reaction rate will become a limiting factor. Comparing PETase with other PET-hydrolyzing enzymes revealed that the PETase from the *Ideonella sakaiensis* has the highest catalytic efficiency for PET degradation at room temperature (a complete degradation of the PET film was observed after 6 weeks at 30 °C).¹¹ This enzyme is thus of considerable interest to develop an innovative green technology to biodegrade PET. From this discovery, some studies have been performed to characterize PETase and discover its mechanisms of PET degrada-

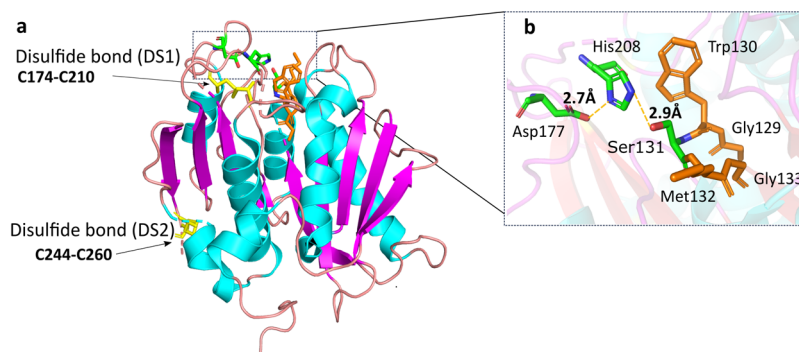


Figure 1. Structure of PETase (PDB ID: 5XG0). a. The crystal structure of PETase is represented as a cartoon, with β -strands in magenta and α -helices in cyan. The residues that form the catalytic triad are represented as green sticks. The disulfide bonds are shown as yellow sticks. The Gly-X-Ser-X-Gly motif is shown as orange sticks. b. Close up view of the catalytic triad, with the critical interaction distances given in Å. The catalytic residues are colored by element. The remaining Gly-X-Ser-X-Gly motif residues are shown in orange.

tion.^{13,14,16–19} Understanding it is a fundamental requisite for the rational engineering of PETases into catalysts for large-scale industrial applications.

1.1. Structure of PETase and PETase Complexes.

Several X-ray crystal structures of PETase can be found in the literature.^{12–14,16,17} PETase (or PET hydrolase, EC 3.1.1.101) is a serine esterase that catalyzes the hydrolysis of esters. Organized in an α/β -hydrolase fold formed by nine β -strands and seven α -helices (Figure 1), PETase contains the serine hydrolase motif Gly-X-Ser-X-Gly, like other α/β -hydrolases, such as lipases or esterases. The enzyme possesses a canonical catalytic triad consisting of Ser-His-Asp, as predicted from sequence alignment with homologous enzymes that exhibit PET-hydrolytic activity.^{12–14,16,17} Han *et al.*¹⁷ identified the PETase catalytic triad as Ser131-His208-Asp177 (numbered as Ser160-His237-Asp206 in Joo *et al.*¹⁴). Site-directed mutagenesis experiments validated the identity of the catalytic triad by characterizing alanine variants that showed very low activity.

PETase has two disulfide bonds (DS1 and DS2), while other homologous enzymes have only one. DS1 (Cys174-Cys210), located near the active site connecting the β 7- α 5 and β 8- α 6 loops, is specific to PETase. Despite stabilizing the enzyme structure, it is assumed that it still keeps the active site flexible enough to compensate for the substrate rigidity.¹⁶ Mutagenesis studies¹⁴ have shown that the additional disulfide bond, DS2 (Cys244-Cys260), connecting the C-terminal helix with the last loop, increases the thermal stability of the enzyme.

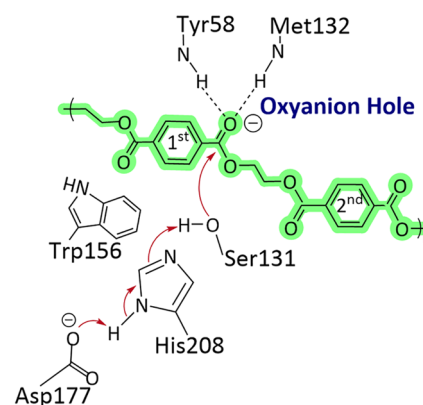
Han *et al.*¹⁷ solved the structure of the R103G/S131A double mutant in complex with 1-(2-hydroxyethyl) 4-methyl terephthalate (HEMT) and *p*-nitrophenol (pNP), obtaining two enzyme/ligand complexes. In the former, the ligand was bound in a groove comprising Tyr58, Trp130, Ala131, Met132, Trp156, Ile179, and His208. Trp156 has an essential role in substrate binding, stabilizing the substrate through π -stacking interactions, whereas the other residues provided labile hydrophobic interactions with HEMT. The backbone NH groups of Tyr58 and Met132 formed hydrogen bonds with the carbonyl of the HEMT-ester, resembling an oxyanion hole arrangement.

An X-ray structure for a PETase:PET complex is not available. Joo *et al.*¹⁴ performed docking calculations with 2-hydroxyethyl-(mono hydroxyethyl terephthalate)₄, 2HE-(MHET)₄, composed of four MHET units. The calculations identified a binding cleft of approximately 40 Å, divided into two binding subsites, I and II. Substrate binding to subsite I

was achieved through π - π interactions between Trp156 and the first phenyl of MHET, and with residues Met132 and Ile 179 helping by providing a hydrophobic surface at the bottom of the subsite. Subsite II was more superficial and accommodated the remainder of the MHET through hydrophobic interactions.

1.2. Proposed Reaction Mechanism of PETase. Han *et al.*¹⁷ showed that PETase followed the classical mechanism of cutinases,²⁰ comprising two stages: acylation, where an acyl-enzyme intermediate was formed, and deacylation, in which the covalent intermediate was resolved and the enzyme returned to its initial state. As shown in Scheme 3, at the

Scheme 3. Reaction Mechanism of PETase Proposed by Han *et al.*^{17a}



^aThe catalytic Ser131 was activated by transferring a proton to His208, generating a nucleophilic alkoxide that attacked the carbonyl group of the substrate; the oxyanion was stabilized by hydrogen bonds provided by an oxyanion hole formed by the peptide amides of Tyr58 and Met132 with the formed acyl-enzyme intermediate. This intermediate was further hydrolyzed to generate the product via a second tetrahedral intermediate, in the deacylation stage. The product was released, and the enzyme was regenerated.

first stage, the Ser131, assisted by His208 and Asp177, carried out the nucleophilic attack to the carbonyl group of the first benzene ring of the substrate; an oxyanion hole formed by the backbone of Tyr58 and Met132, which polarized the carbonyl oxygen of the ester bond of the substrate and stabilized an acyl-enzyme reaction intermediate. The latter was subsequently

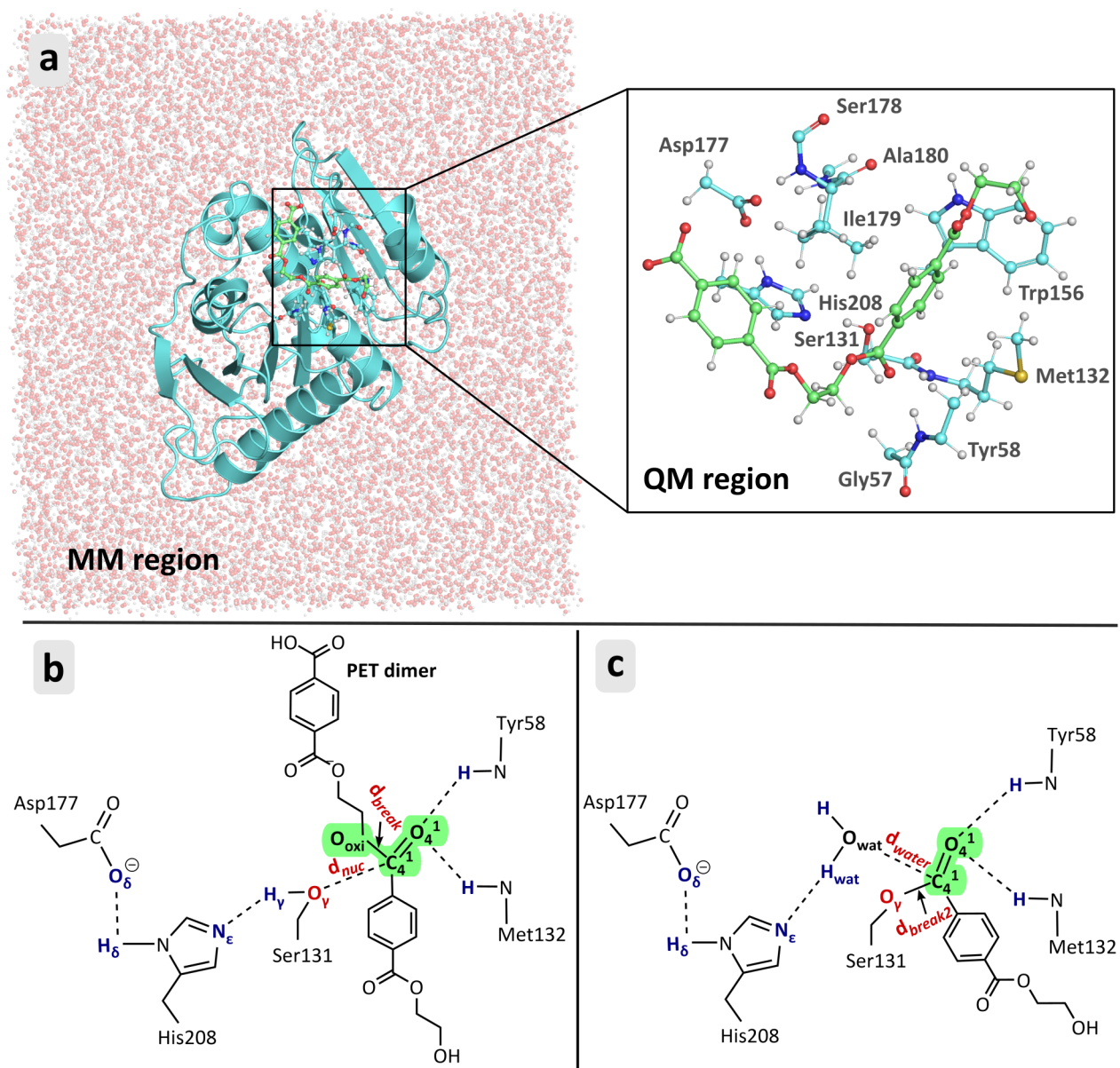


Figure 2. a. Left: The system that was used in the simulations; the inset on the right represents the QM region of the model. b. 2D representation of the QM atoms and interaction distances at the reactant state of the acylation step. c. 2D representation of the QM atoms and interaction distances at the reactant state of the deacylation step.

resolved in a nucleophilic attack by a water molecule, during the second reaction stage, restoring the enzyme to its initial state. After the ester bond was cleaved, the π -stacking interaction between the benzoic group and the Trp156 rotated, and the product was released from the active center.

Boneta *et al.*¹⁹ described the chemical mechanism of PETase in four reaction steps, a two-step acylation and a two-step deacylation, each mediated by tetrahedral enzyme–substrate intermediates, using semiempirical AM1/MM umbrella sampling calculations with a posteriori corrections at the DFT (M06-2X) level, and further characterized the geometry of the four transition states at the M06-2X/6-31+G(d,p):MM level. In a previous work of Knott *et al.*²¹ for MHETase, where they carried out QM/MM simulations at the SCC-DFTB:MM level, the authors suggested that the reaction occurred in two steps without the formation of stable tetrahedral intermediates. An alternative approach by Pinto *et al.*,²² also for MHETase, at the

B3LYP/GPW:MM level, indicated that the reaction mechanism obtained is analogous to that of canonical serine hydrolases, where the acylation and deacylation steps occur via a metastable tetrahedral intermediate. Here, we used a robust QM/MM umbrella sampling method, at the density functional theory level (PBE) with a larger QM region (146 atoms), and identified all its intermediates and transition states with high accuracy. Our results support a two-step mechanism consistent with that proposed by Han *et al.*,¹⁷ in which transient tetrahedral conformations can be observed as each reaction approaches the corresponding transition state.

Despite the very high cost of our calculations, the superior accuracy they provided was necessary to predict enzyme mutations that increased the enzyme turnover, which we did in the second part of this work. Based on the electronic structure of the rate-limiting step, we put forward the rationale for rate-enhancing enzyme engineering and proposed specific enzyme

mutants that should increase the enzyme reaction rate. This study provides new insights on the rational engineering of PETase for more efficient biodegradation of PET waste.

2. METHODS

2.1. Modeling the PETase:Substrate Complex. A PETase:substrate model was prepared from the crystal structure with PDB ID: SXH3 at 1.30 Å resolution,¹⁷ which has three ligands, one of them in the active site, and two mutations (Arg103Gly and Ser131Ala). The PyMOL software²³ was used to revert the mutated residues with the mutagenesis tool. In previous studies, different types of substrates have been used as PET analogs to determine the mechanism of molecular degradation. Han and co-workers¹⁷ used two ligands in their study, HEMT and pNP (Figure S1); Joo *et al.*¹⁴ used 2-hydroxyethyl-(monohydroxyethyl terephthalate)₄, 2-HE(MHET)₄, in their work (structures in Figure S1); Fecker *et al.*¹⁶ used PETase in complex with a PET dimer. We modeled a PET dimer from the HEMT ligand present at the active site of PETase in the structure with the PDB ID SXH3, as the sp² ester-carbon was near the Ser131 (2.3 Å) and kept the conserved terephthalate moiety.

The algorithms for pK_a prediction in proteins are fallible, due to the very complex electrostatic environment of these large molecules,^{24,25} and consequently the protonation state of the residues of PETase was assigned after careful visual inspection of the environment of each residue and considering the pK_a values obtained by PROPKA 3.0^{26,27} for the reference physiological pH (7.0). As a result, residues His75 (predicted pK_a of 3.29) and His208 (predicted pK_a of 5.29) were protonated at the δ-nitrogen, according to the most likely network of hydrogen bonds formed with neighboring residues/solvent molecules. The geometry of the PET dimer substrate was optimized with the Gaussian 09 software²⁸ at the HF/6-31G(d) level to determine RESP²⁹ atomic point charges. The antechamber module of the Amber 18 software package was employed to parametrize intramolecular bonds and Lennard-Jones parameters of the substrate using the general Amber force field (GAFF2).³⁰ The tleap module from Amber 18³¹ was used to assemble the system, using the ff14SB³² force field to parametrize the enzyme and abovementioned parameters for the substrate. The total charge of the system was +6, neutralized with six chlorine ions. The enzyme/substrate complex was centered in a rectangular water box with faces at a minimum distance of 12 Å from the limits of the enzyme:substrate complex and solvated with TIP3P³³ water molecules. The final system had 34 821 atoms.

The simulations were performed using the sander module of the Amber 18 package. The system's geometry was first optimized in five sequential steps (details in the SI), followed by MD simulations. In the latter, all bond lengths involving hydrogen atoms were restricted with the SHAKE algorithm.³⁴ The time step for the integration of the equations of motion was 2 fs. The particle mesh Ewald (PME) method³⁵ with a cutoff distance of 10 Å was used to calculate the Coulomb interactions. A linear heating stage (annealing) from 0 to 300 K was performed for 40 ps, followed by 60 ps at 300 K in the NVT ensemble. We used the NPT ensemble at 300 K and 1 bar for further equilibration and production, controlled by the Langevin thermostat³⁶ and Berendsen barostat.³⁷ A first 3.5 ns MD was performed with restraints on the enzyme and ligands, followed by a production of 50 ns with no restraints. A root-mean-square deviation (RMSD) analysis of the enzyme's

backbone was performed to follow the system's stability during the MD (Figure S2). Interatomic distances of catalytic atoms were also observed throughout the simulation time.

2.2. QM/MM MD Simulations. Born–Oppenheimer molecular dynamics (BOMD) was performed using the software package CP2K.³⁸ Density functional theory (DFT) was used for the QM calculations. Despite its high computational cost, it provides the best accuracy possible nowadays within an umbrella sampling (US) protocol,^{39,40} the PBE^{41,42} functional was used with a double-ζ valence polarization plane-wave basis-set (DZVP-GTH-PBE) and Goedecker-Teter-Hutter (GTH)^{43,44} pseudopotentials. The cutoff for the plane-wave expansion was set to 300 Ry in a QM cell, with dimensions 26.14 Å × 24.91 Å × 24.14 Å. The QM cell included the Ser131, side chain and part of the backbone of Met132, backbone, and part of the side chain Tyr58, part of the backbone of Gly57 and Ala180, the PET dimer substrate, and the side chain of the Trp156, Asp177, Ser178, Ile179, and His208 (Figure 2a); all comprising a total of 146 atoms, a charge of −2 and a singlet spin multiplicity. The remaining 34 675 atoms were assigned to the MM part and described with the Amber ff14SB force-field. The boundary region was treated with hydrogen atoms that were used as “link atoms”.^{45,46} The long-range Coulomb behavior was described by the Gaussian expansion of the electrostatic potential method (GEEP).³⁸

According to the RMSD of the active site residues, we clustered the MD trajectories for the acylation reaction and started with a representative structure from the most populated cluster (Table S1 and Figure S3). No water molecules were observed in the vicinity of the catalytic residues Ser131 and His208, neither in the representative structure of the cluster nor in the structures composing the cluster. A similar analysis was carried out for the model of the deacylation reaction, where we found water molecules close to Ser131 and His208, which were considered for the construction of the QM region. Both models were optimized before the QM/MM MD simulations, using electrostatic embedding. The optimized structures were equilibrated with MD for 2 ps in the NVT ensemble, at a temperature of 300 K using the CSVR thermostat,³⁸ and an integration step of 1 fs. Subsequently, QM/MM steered MD was performed with a harmonic potential force constant of 50 kcal·mol^{−1}·Å^{−2} and a target growth of 0.002 Å·fs^{−1}, for 3 ps along a tentative reaction coordinate for both the acylation and deacylation stages.

The following reaction coordinates (RC in the equations) were used: (i) RC_{acyl} = $d_{\text{break}} - d_{\text{nuc}}$ for the acylation step (Figure 2b), where d_{nuc} represents the nucleophilic attack of the Ser131-O_γ to the carbonyl carbon of the PET dimer (C₄¹) and d_{break} represents the breaking of the PET dimer-ester bond, C₄¹-O_{oxi}; (ii) RC_{deacyl} = $d_{\text{break2}} - d_{\text{water}}$ for the deacylation step (Figure 2c), where d_{break2} represents the breaking of the O_γ-C₄¹ bond of the acylated serine and d_{water} represents the nucleophilic attack of an active-site water oxygen (O_{wat}) to C₄¹.

2.3. Umbrella Sampling Simulations. We calculated the free energy profile along the reaction coordinates using umbrella sampling simulations.⁴⁷ The initial coordinates for the US windows, linearly spaced by 0.1 Å, were taken from the steered MD simulations. The harmonic potentials for the windows were 50 or 100 kcal·mol^{−1}·Å^{−2} to obtain a good overlap of the US windows (Figure S4 and S5). All US simulations ran for 15 ps, in each of the 47 windows for the first reaction step and the 44 windows for the second reaction

Table 1. Average Distances and Angles Obtained for the Reactants, Transition State, and Intermediate of the First Step of the Reaction^a

Distances	R	TS1	INT1
Ser131 O _γ –C ₄ ¹	3.30 ± 0.14	1.49 ± 0.05	1.37 ± 0.04
Ser131 OH _γ –His208 N _ε	1.76 ± 0.15	1.26 ± 0.26	1.99 ± 0.28
His208 NH _δ –Asp177 O _δ	1.62 ± 0.15	1.39 ± 0.24	1.63 ± 0.15
Tyr58 NH–O ₄ ¹	2.68 ± 0.57	2.05 ± 0.21	2.25 ± 0.23
Met132 NH–O ₄ ¹	3.07 ± 0.44	2.19 ± 0.26	2.31 ± 0.30
C ₄ ¹ –O _{oxi}	1.37 ± 0.04	1.71 ± 0.15	3.22 ± 0.11
Ser131 O _γ –H _γ	1.02 ± 0.04	2.15 ± 0.42	3.18 ± 0.27
O _{oxi} –Ser131 OH _γ	3.78 ± 0.29	1.84 ± 0.69	1.01 ± 0.04
Angles	R	TS1	INT1
Tyr58 N–H–O ₄ ¹	139.10 ± 20.95	160.01 ± 12.50	158.91 ± 12.74
Met132 N–H–O ₄ ¹	154.10 ± 13.77	162.65 ± 10.16	149.40 ± 14.03

^aDistances are represented in Å as average ± standard deviation. Angles are represented in degrees as average ± standard deviation. Substrate atom names are given in Figure 2b.

step. The total sampling was 0.7 ns. The weighted histogram analysis method (WHAM)⁴⁸ was used to recover the unbiased free energy from the umbrella sampling simulations, using 100 bootstrap data sets and a 0.0001 tolerance threshold along a number of bins twice the number of independent simulated windows. Even though the resulting statistical errors are minimal (0.02–0.07 kcal·mol^{−1}), the energies are presented in kcal·mol^{−1} with one decimal place because the error of the PBE/AMBER method is ~3 kcal·mol^{−1}. We present the inter- and intramolecular distances to two decimal places (pm), which is the typical accuracy of the PBE functional.

3. RESULTS AND DISCUSSION

3.1. MD Simulations. The 50 ns MD simulation of the PETase/PET dimer complex was monitored to verify the structural stability of the system and the interactions of the residues involved in the enzymatic reaction. The backbone root-mean-square deviation (RMSD) for the protein and active site residues was stable throughout the whole 50 ns simulation, with an average RMSD of 0.75 ± 0.07 Å for the protein backbone and 0.56 ± 0.04 Å for the active site. (Figure S2). Visual inspection of PETase and the PET dimer interactions showed that residues Thr59, Ala60, Trp130, Trp156, Ile179, Ser 207, and Ser209 seem to contribute to the binding of the polymer to the enzyme (Figure S6). These interactions are mostly of van der Waals type, with interactions between aromatic moieties and other hydrophobic contacts preserved for most of the MD simulations. Considering both units of the PET dimer, the one binding the active site is more rigid. In addition, dispersive interactions are more frequent when the residues composing the oxyanion hole with the C₄-carbonyl of PET dimer are farther apart.

3.2. Study of the Reaction Mechanism. We studied the reaction mechanism proposed for the serine hydrolases by simulating along the difference between the breaking and forming of the C–O bonds at both steps of the reaction (represented in Figure 2b as $d_{\text{break}}/d_{\text{nuc}}$ and in Figure 2c as $d_{\text{break2}}/d_{\text{water}}$). Such a representation to map the free energy of the reaction has been used in other studies of hydrolases^{21,49} and can evaluate both the synchronicity of each reaction step and the formation of a tetrahedral intermediate, while it does not predetermine the base-assisted mechanism.

3.2.1. Acylation Reaction. In the reactant state, the nucleophilic Ser131 is close to C₄¹ (Ser131 O_γ–C₄¹ distance of 3.30 ± 0.14 Å), and the hydrogen bonds provided by the

oxyanion hole constituted by Tyr58 and Met132 are marginally formed (2.68 ± 0.57 and 3.07 ± 0.44 Å). The distances of the fundamental interactions for the acylation step are presented in Table 1.

Umbrella sampling simulations along the reaction coordinate generated a free energy profile for the acylation stage (Figure 3d), which took place through a single, concerted step, as it encompassed a single transition state (TS1) at RC_{acyl} = +0.2 Å. The free energy barrier was 20.0 kcal·mol^{−1}, in agreement with that determined experimentally for BHET (18.0–18.6 kcal·mol^{−1}).^{11,50} The reaction free energy was 4.7 kcal·mol^{−1}.

The reaction started with the deprotonation of Ser131 by His208, followed by a nucleophilic attack of Ser131-O_γ to C₄¹, leading to a tetrahedral transition state. In the path from the transition state to the products, the ester bond of the substrate broke, and the leaving MHET deprotonated His208, ending up with the formation of the acylated PETase intermediate.

Figure 3 represents the structural details of the reactant R, the transition state TS1, and the first intermediate INT1 state. At TS1, the Ser131 H_γ–O_γ distance was 2.15 ± 0.42 Å and the Ser131 O_γ–C₄¹ distance was 1.49 ± 0.05, confirming that the Ser131 was fully deprotonated and already bound to the substrate. A tetrahedral transition state emerged, as the substrate's scissile bond C₄¹–O_{oxi} was stretched but still formed (1.71 ± 0.15 Å). The distance between the Ser131 hydroxyl proton and the His208 N_ε atom was 1.26 ± 0.26 Å at TS1. Proton transfer to His208 (1.16 ± 0.14 Å) was achieved on the path toward TS1 (Figure S7 and Table S2). However, this bond stretched in the region close to TS1 because the proton was quickly displaced toward the oxygen atom O_{oxi}. This phenomenon was provoked by the negative charge growing at O_{oxi} as the C₄¹–O_{oxi} bond of the substrate was cleaved. In support of this interpretation, the distance from the Ser131 hydroxyl proton to the oxygen O_{oxi} decreased to 1.84 ± 0.69 Å at TS1.

The cleavage of the PET dimer ester bond took place in the decay from TS1 to INT1, with the O_{oxi}–C₄¹ distance increasing from 1.71 ± 0.15 to 3.22 ± 0.11 Å. The leaving MHET captured the Ser131-H_γ proton, as the distance from the oxygen O_{oxi} to the Ser131-H_γ changed from 1.84 ± 0.69 Å at TS1 to 1.01 ± 0.04 Å at INT1.

The acylation reaction took place through a concerted but asynchronous path; the nucleophilic attack occurred after the deprotonation of the Ser131 by His208, and the breaking of

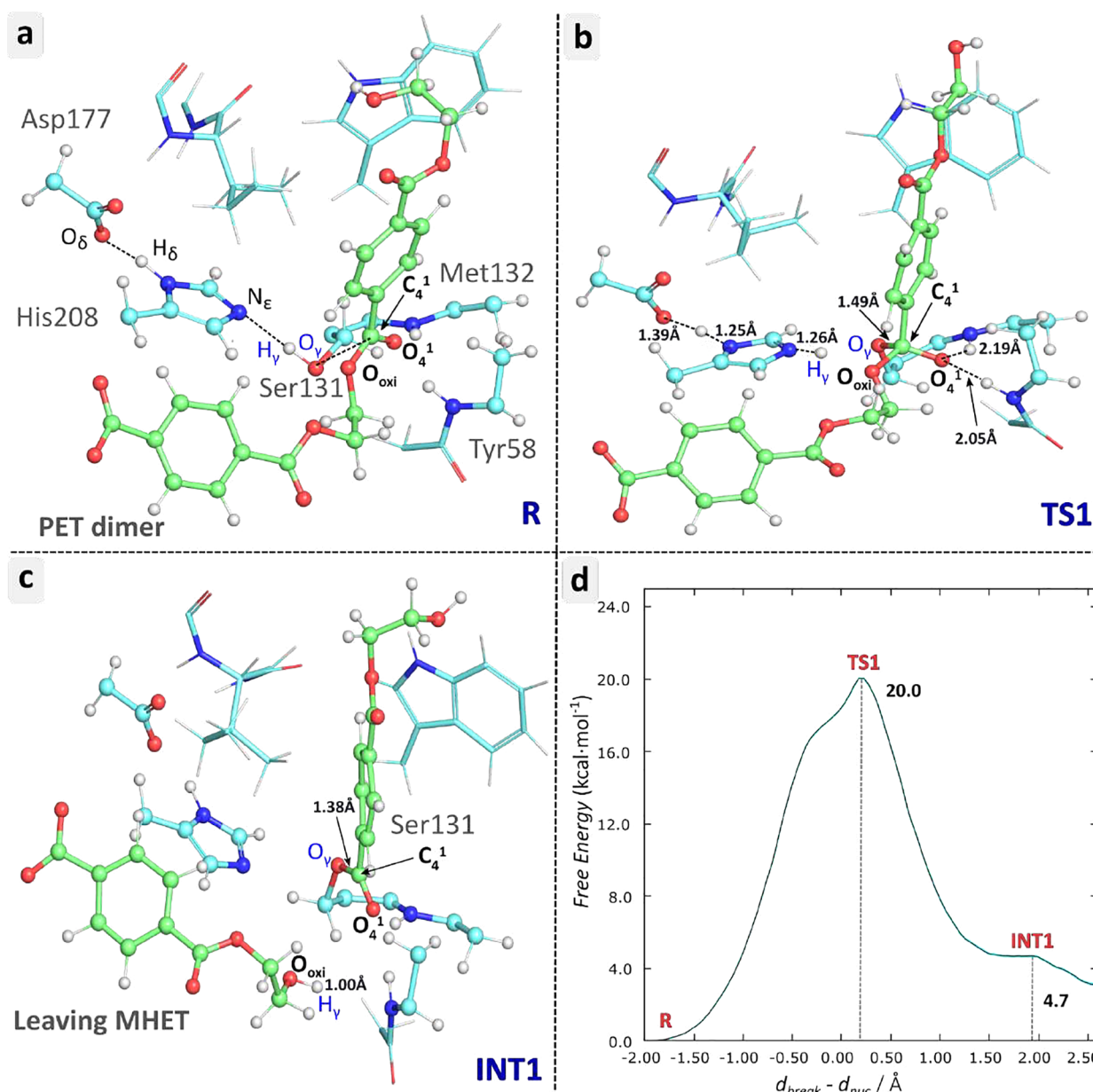


Figure 3. Acylation reaction: panel a Ser131 was deprotonated by His208 and performed a nucleophilic attack to C_4^1 with the formation of a tetrahedral transition state TS1 (panel b). During TS1 decay to INT1, O_{oxi} deprotonated His208 and the $C_4^1-O_{oxi}$ bond was cleaved, releasing MHET, which left from the active site (panel c). The free energy profile for the acylation step is represented in panel d (average free energy in black, $\pm$ one standard deviation in light blue).

the ester bond occurred after the nucleophilic attack of Ser131. The protonation of the MHET leaving group was the latest event.

In agreement with experiments^{51–54} where the oxyanion hole contributed for the TS stabilization through the formation of hydrogen bonds that are fundamental to perform the catalytic mechanism, the reaction was favored by the oxyanion hole formed by the backbone amine of Tyr58 and Met132. To elucidate the contribution of the oxyanion hole in the first step of the reaction, we analyzed the distances and angles of the atoms of the backbone-NH of Tyr58 and Met132 with the O_4^1 atom. Tables 1 and S2 provide a comparison between these distances and angles. While the distances from the backbone-NH of Tyr58 and Met132 to the O_4^1 atom shortened from R to TS1 (2.68 ± 0.57 to 2.05 ± 0.21 Å and 3.07 ± 0.44 to 2.19 ± 0.26 Å), the interaction angles became more linear (160.01

$\pm 12.50^\circ$ and $162.65 \pm 10.16^\circ$), indicating that these hydrogen bonds became tighter as a result of building negative charge at the O_4^1 atom, and confirming the effect of the oxyanion hole at stabilizing the transition state.

Upon formation of INT1, a clear minimum could not be observed (refer to Figure S7), and the free energy decreased as the reaction coordinate increased. However, a closer look at the main interacting distances at the active site shows that only those involving the MHET leaving group and the oxyanion hole at O_4^1 are increasing significantly (refer to INT1, IP4, and IP5 of Table S2). Hence, we believe that the diffusion of INT1 to the bulk solvent is occurring after INT1.

In fact, we carried out conventional MD simulations of INT1, where we observed that the MHET molecule formed was diffused to the solvent on the nanosecond time scale (in fact, in less than 1 ns), being replaced by water molecules from

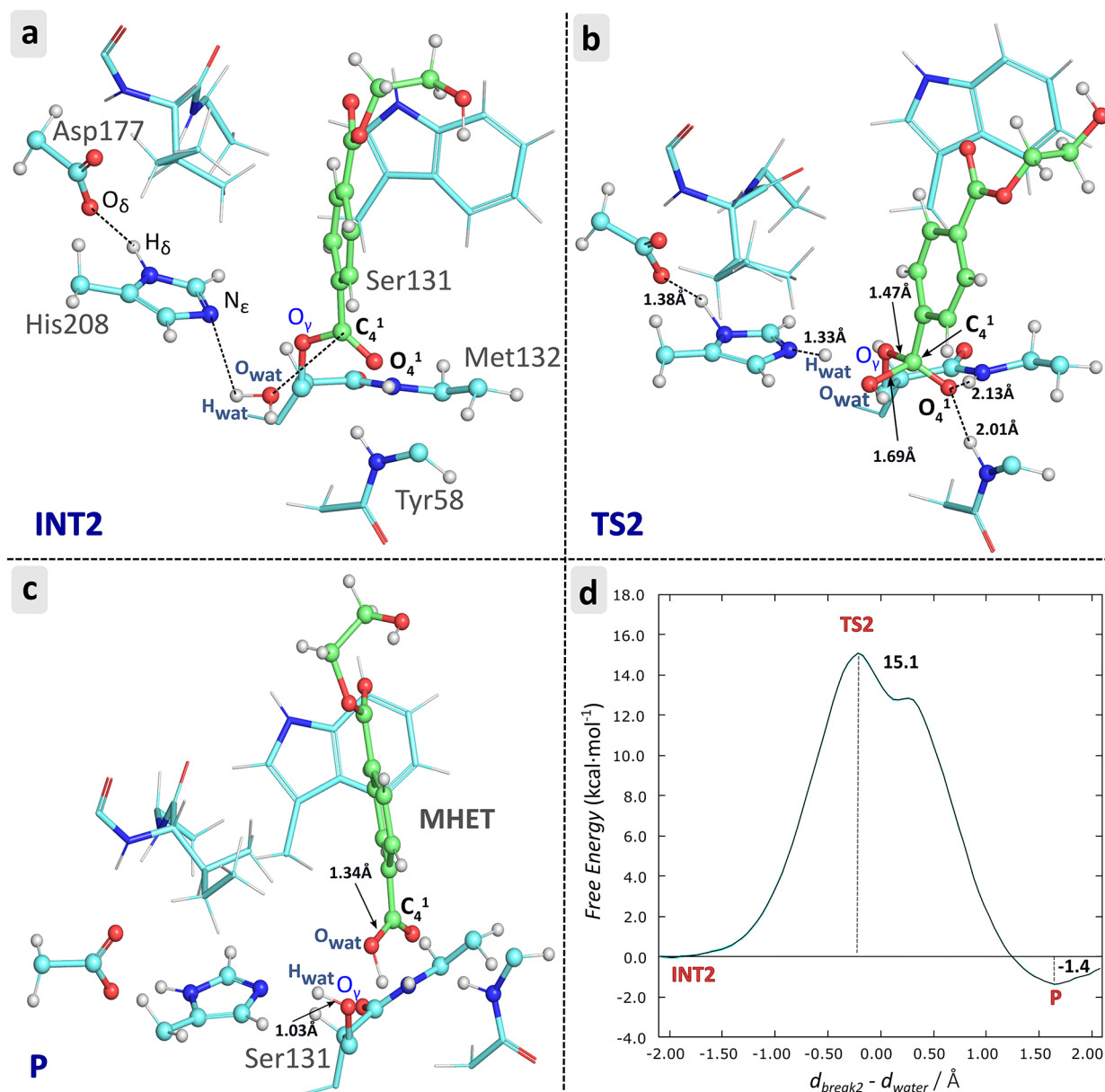


Figure 4. Deacylation reaction: (panel a) His208 deprotonated the water molecule, which performed a nucleophilic attack on C_4^1 , as shown in TS2 (panel b). During the TS2 decay to P, the substrate-Ser131 bond broke, and Ser131 deprotonated His208, restoring the catalytic serine and generating a second MHET molecule (panel c). The free energy profile for the acylation step is represented in panel d (average free energy in black, $\pm$ one standard deviation in light blue).

the bulk solvent. Thus, the MD simulation suggests that MHET diffusion, leading to the INT2 state, should be a spontaneous entropy-driven process, and that, given the slightly endergonic acylation step and the decreasing free energy after INT1, the overall process should be exergonic.

3.2.2. Deacylation. In the deacylation step, an active-site water molecule occupying the region left available after the diffusion of the leaving MHET to the bulk solvent, at the end of the acylation step, performed a nucleophilic attack on the enzyme–substrate adduct, generating the final product and restoring the resting state of the enzyme. The initial deacylation state (INT2) was similar to INT1, but MHET had already left the active site. Water molecules were abundant at the active site, as seen in the MD simulations. One of them took part in the reaction (Figure 4a).

The initial distance between His208 N_ϵ and the water molecule (2.49 ± 0.92 Å) was favorable for the water deprotonation, and the distance of the water molecule to C_4^1 (3.27 ± 0.12 Å) also favored the nucleophilic attack (Table 2). The hydrogen bond between Asp177 and His208 was stable (1.70 ± 0.14 Å).

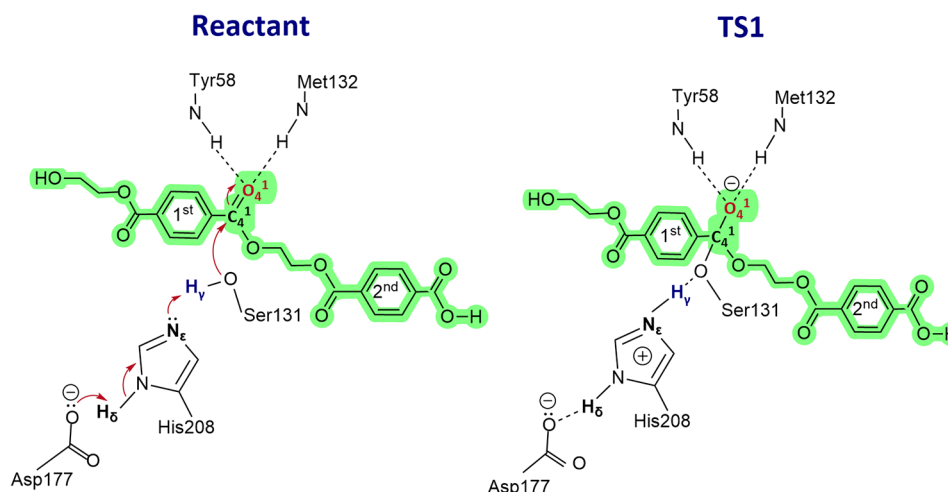
Figure 4 shows the free energy profile of the deacylation process, in which three relevant states were observed: the reactant (INT2), the transition state (TS2), and the products of this step and enzyme cycle (P). The reaction had an activation free energy of 15.1 kcal·mol $^{-1}$ and a reaction free energy of -1.4 kcal·mol $^{-1}$. Contrary to the acylation step, the deacylation step exhibits clear extrema, which we infer to relate to the tighter binding of the bonded MHET to the active site of PETase.

Table 2. Average Distances Obtained for the Reactants, Transition State, and Intermediate of the First Step of the Reaction^a

Distances	INT2	TS2	P
Ser131 O _γ –C ₄ ¹	1.37 ± 0.03	1.47 ± 0.05	3.04 ± 0.11
O _{wat} –C ₄ ¹	3.27 ± 0.12	1.69 ± 0.13	1.34 ± 0.03
H _{wat} –His208 N _ε	2.49 ± 0.92	1.33 ± 0.28	1.69 ± 0.13
H _{wat} –Ser131 O _γ	3.15 ± 0.42	2.42 ± 0.30	1.03 ± 0.04
NH _δ –O _δ	1.70 ± 0.14	1.38 ± 0.32	1.58 ± 0.27
Tyr58 NH–O ₄ ¹	2.24 ± 0.25	2.01 ± 0.18	2.10 ± 0.23
Met132 NH–O ₄ ¹	2.38 ± 0.41	2.13 ± 0.23	2.68 ± 0.33
H _{wat} –O _{wat}	1.00 ± 0.03	1.46 ± 0.46	3.71 ± 0.30
Angles	INT2	TS2	P
Tyr58 N–H–O ₄ ¹	159.60 ± 12.50	164.42 ± 8.53	158.90 ± 12.60
Met132 N–H–O ₄ ¹	148.53 ± 15.13	162.50 ± 10.52	156.67 ± 13.40

^aDistances are represented in Å as average ± standard deviation. Angles are represented in degrees as average ± standard deviation. Substrate atom names are given in Figure 2c.

Scheme 4. Rate-Limiting Step of the PETase Catalytic Mechanism Involves a Substantial Transformation in the Charge Distribution of the System^a



^aAt TS1, a negative charge is located on the carbonyl oxygen O₄¹ and a positive charge is located on the His208 imidazole, creating a charge separation that was absent in the reactants when both species were neutral.

At TS2, the Ser131 O_γ–C₄¹ distance was 1.47 ± 0.05 Å, which corresponded to a still incipient elongation of the Ser-acyl C–O bond. The distance from the water oxygen to C₄¹ was 1.69 ± 0.13 Å. Such distances corresponded to a clear tetrahedral transition state, similar to TS1. Unlike at the acylation step, the nucleophilic water attack was synchronous with its deprotonation by His208, as the distance between the water hydrogen and the His208 N_ε was 1.33 ± 0.28 Å, and the water O–H bond was stretched to 1.46 ± 0.46 Å.

The oxyanion hole hydrogen bonds got shortened from INT2 to TS2: 2.24 ± 0.25 to 2.01 ± 0.18 Å (Tyr58) and 2.38 ± 0.41 to 2.13 ± 0.23 Å (Met132), confirming that this interaction stabilized the oxyanion formed at C₄¹ at TS2, as was observed for the acylation step. However, the angles involving the backbone-NH of Tyr58 and Met132, and the O₄¹, were mostly conserved across the step, which indicates that the contribution of the oxyanion hole may play a larger stabilizing role during the acylation step.

In the products, the Ser131 O_γ–C₄¹ bond was definitively broken (3.04 ± 0.11 Å). Ser131 deprotonated the His208 in the path from TS2 to P, and a neutral Ser131 was regenerated (water proton–Ser131 O_γ distance of 1.03 ± 0.04 Å). The bond between the water oxygen O_{wat} and the substrate carbon

atom C₄¹ shortened to 1.34 ± 0.03 Å, confirming the formation of the product MHET.

3.3. Engineering of PETase to Increase the Reaction Rate. The engineering of PETase is fundamental for its successful implementation in large-scale recycling. Two factors can be promoted by engineering, *i.e.*, enzyme thermal stability and efficiency.

The first helps PETase to work at higher temperatures, where the fraction of recalcitrant crystalline PET is smaller or nonexistent. Previous research of da Costa *et al.*,⁵⁵ have used molecular docking and MD simulations to analyze structural changes of PETase induced by PET. They demonstrated that the β1–β2 connecting loop is very flexible and can be considered for mutagenesis to increase the thermal stability of PETase. Although the thermal stability is a relevant problem, it falls out of the scope of this work.

The second factor, *i.e.*, enzyme efficiency, fits our purpose well, as the free energy profile and the identification of the structure of the rate-determining step accomplished here provide the basis for the rational engineering of enzyme efficiency. Enzyme engineering can be improved by decreasing *K_M* as, for the design of new PETases, improving the enzyme–PET complex formation step can be crucial.⁵⁵ Increasing *k_{cat}* is

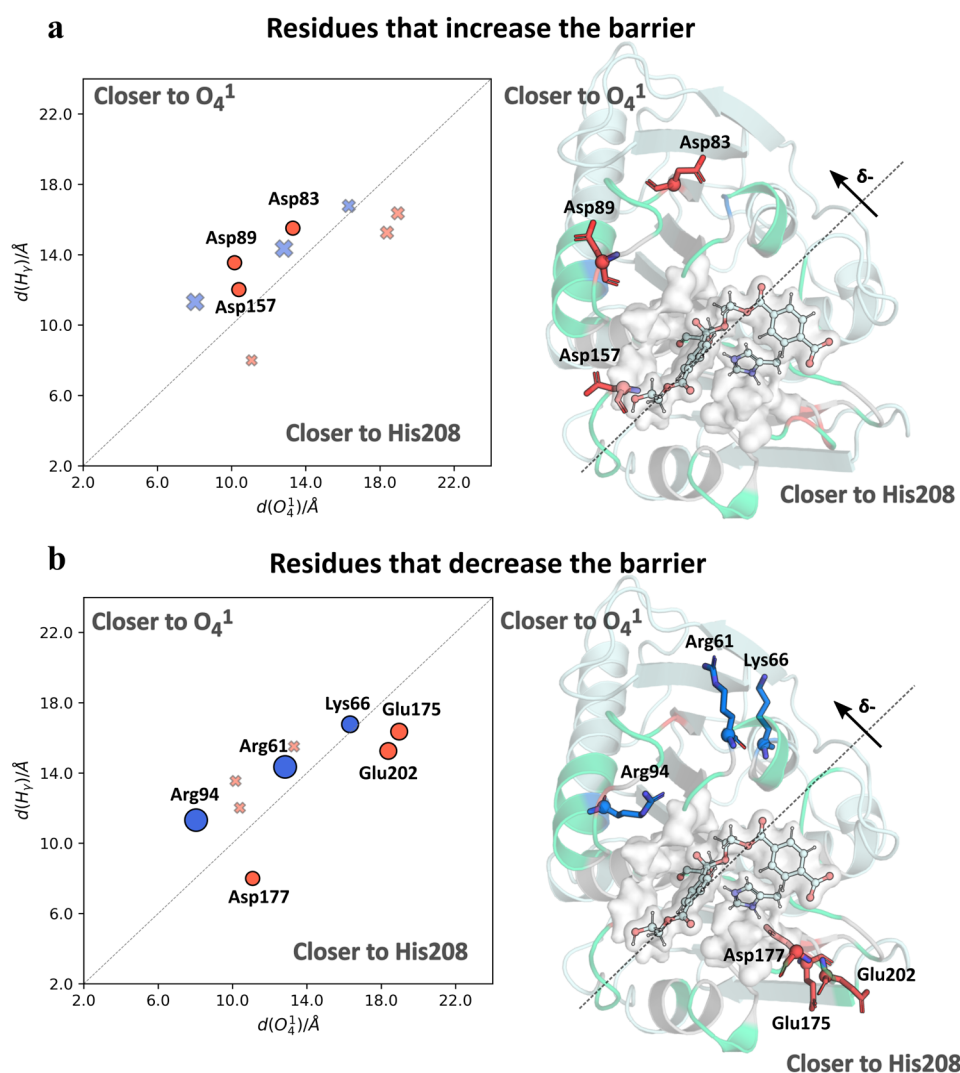


Figure 5. Panel a: Charged residues that increase the barrier. (Left) Distance of charged residues within 10 Å from the active site to the O_4^1 atom and the His208 H_{ϵ} atom. Three Asp residues closer to the O_4^1 atom than His208 raise the barrier. (Right) The three Asp residues that raise the barrier, shown in sticks within the PETase/PET dimer complex shown in cartoon and ball-and-stick representation. Panel b: Charged residues that decrease the barrier. (Left) Distance of charged residues within 10 Å from the active site to the O_4^1 and to the His208 H_{ϵ} atoms. (Right) Shown in sticks within the PETase/PET dimer complex shown in cartoon and ball-and-stick representation. The C_{γ}/C_{δ} atoms of Asp/Glu, the N_{ϵ} atom of Lys, and the C_{ϵ} of Arg were used in the distance measurement to represent the carboxylate, amino, and guanidinium groups. The dashed line separates the region closer to His208 from the region closer to the O_4^1 atom. The direction in which the negative charge flows when moving from the reactant to the transition state is also depicted in both panels.

also a strategy that brings great rate enhancements in enzyme engineering. It is the most commonly used, and it is the one that will be addressed here.

Scheme 4 shows the charge flow from the reactant state to TS1 (the rate-limiting step) and reveals two zones to consider. The first one, the positively charged zone, corresponds to the protonated His208 imidazole, and the second one is the negatively charged O_4^1 formed in the nucleophilic attack of Ser131 to the PET dimer.

We identified the charged residues having at least an atom within 10 Å of the active site and measured the distance between their negative/positive zones to His208 (specifically its H_{ϵ} , as the salt-bridge with Asp177 shields the interactions with the H_{δ}) and oxygen O_4^1 . The measurements were made at R and TS1 states. The contribution of each of these residues to the increase/decrease of the barrier is explained by the differential stabilization exerted upon R and TS1 (the essence

of catalysis): (i) if a positive residue is closer to O_4^1 than to the imidazole nitrogen atoms it will stabilize TS1 more than R and lower the activation barrier; if it is closer to the imidazole nitrogen than to O_4^1 it stabilizes more R than TS1, causing an activation energy increase; (ii) the same reasoning applies to negative residues: these lower the barrier if they are closer to the His208 imidazole than to O_4^1 and raise the barrier if they are closer to O_4^1 than to the His208 imidazole.

The analysis shows more charged residues closer to the O_4^1 atom than to His208 (Figure 5). More charged residues stabilize TS1 in relation to R (six) than the contrary (three), as expected from a catalyst. The negative residues Glu175, Asp177, and Glu202 are closer to His208 than to the O_4^1 atom and are expected to decrease the barrier of the rate-limiting step, while Arg94, Arg61, and Lys66 closer to the O_4^1 atom than His208, are also expected to lower the barrier. As a result, these residues should not be mutated. The negative Asp83,

Asp89, and Asp157, closer to the O_4^1 atom, are thus expected to increase the barrier, and could be candidates for mutagenesis by neutral or positive residues.

The representation of the protein in panels a (right) and b (right) in Figure 5 shows the distribution of the residues that may increase/decrease the activation energy of the rate-limiting step. We suggest that residues Asp83, Asp89, and Asp157 might be mutated by neutral or positively charged residues in order to stabilize TS1 more than R. In contrast, the residues Arg61, Lys66, Arg94, Glu175, Asp177, and Glu202 should be kept unchanged in the structure since they would help stabilize the charges of the polarized zones in the transition state.

Mutations are often deleterious for the enzyme structural stability.⁵⁶ Hence, the mutations proposed here need to overcome this obstacle, something that is never guaranteed. As the proposed mutations are located in flexible loops far from the binding and active sites, they should not be highly destabilizing. Hence, our proposed mutations may reduce the free energy barrier of the rate-limiting acylation step of PETase while retaining the enzyme fold and substrate binding. Alternatively, compensatory mutations can be introduced to mitigate the risk of destabilizing the PETase structure.

4. CONCLUSIONS

Our study applies a robust level of theory (PBE:AMBER) combined with umbrella sampling to describe the catalytic mechanism of PET degradation by PETase, with atomistic and energetic detail, using a model PET dimer as substrate. The results provide support for the general proposal that the mechanism of the acylation of Ser131 is similar to the canonical reaction mechanism of serine hydrolases. Both acylation and deacylation occur through a single, associative, tetrahedral transition state in concerted and asynchronous reaction steps. This proposal is an alternative to the earlier one by Boneta *et al.*, where a four-step mechanism is put forward for PETase, and it is consistent with known experimental data and the characterized reaction mechanism of other serine hydrolases with a similar fold.

Both acylation and deacylation transition states were extensively stabilized by the oxyanion hole from the backbone amines of residues Tyr58 and Met132. The oxyanion hole hydrogen bonds were present in the whole reaction and completely formed in all transition states, providing a catalytic effect. The determined free energy activation barrier for the acylation step was 20.0 and 15.1 kcal·mol⁻¹ for the deacylation step. Both values are consistent with the experimental data (18.0–18.6 kcal·mol⁻¹). Since conventional molecular dynamics simulations showed that the leaving molecule resulting from the acylation reaction diffuses to the bulk solvent at the ns-time scale, being quickly replaced with water molecules from the bulk solvent, the formation of the TS1 is the rate-determining step of the overall reaction mechanism.

An analysis of the charge distribution in the reactants and the rate-limiting transition state allowed us to identify which charged residues have the higher probability of leading to an increase (and to a decrease) of the enzyme's turnover upon mutation by neutral residues. Asp83, Asp89, and Asp157 were identified as potential mutation targets since the results suggest that they would increase the activation barrier, opening the possibility of a rational design study of PETase mutants to make the industrial application of PET biodegradation a viable process.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.1c03700>.

Topology and coordinate files (ZIP)

Description of the geometry optimization steps to relax the structure of the PETase:substrate complex. Figure S1: Compounds used as PETase substrates. Figure S2: RMSD fluctuation of the protein and active site during the 50 ns MD simulation. Figure S3: Snapshot of the representative structure from the most populated cluster used for the construction of the QM/MM model. Figure S4: Density vs umbrella sampling window for the reaction coordinate $d_{\text{break}} - d_{\text{nuc}}$. Figure S5: Density vs umbrella sampling window for the reaction coordinate $d_{\text{break2}} - d_{\text{water}}$. Figure S6: The binding site of the PET dimer. Table S1. Representative distances between catalytic atoms in the cluster. Figure S7: analysis of the gradient of the free energy plot for the acylation reaction. Table S2: Average distances and angles obtained for R, TS1, and INT1 and inflection points for the acylation reaction. Figure S8: analysis of the gradient of the free energy plot for the deacylation reaction. Table S3: Average distances and angles obtained for INT2, TS2, and P and inflection points of the deacylation reaction. (PDF)

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Notes

The authors declare no competing financial interest.

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